Geometric Prediction and Exact-Root Certification of Finite-Error Ordering in Quantum Control

Y. Y. N. Li
Independent Researcher

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Abstract

On a protocol-frozen prospective twenty-path cohort, a zero-point quartic geometric scalar strongly predicts held-out finite-error performance. Separately, on a frozen twelve-path formal cohort, locally unique roots exactly matched in projective output state and first projective response obey a complete 66-pair finite-error ordering certified by outward-rounded interval propagation.

We study this phenomenon in a two-atom Rydberg model with piecewise-constant phase control. Twenty-four control phases are constrained to match the output state and its first-order state response to global amplitude, detuning, and interaction errors. The resulting numerical constraint Jacobian has rank sixteen, leaving an eight-dimensional numerical null space that organizes the candidate implementation fibre. Paths on this fibre display distinct finite-error infidelities.

We first identify a fourth-order symmetric response coefficient, $\mathcal{G}_4$, as a strong low-order predictor of mean finite-error performance. In a protocol-frozen prospective twenty-path validation it attains Spearman correlation 0.996992 and predicts the best path correctly. We then recover the full three-dimensional symmetric fourth-order tensor. Under a nonorthogonal change of error coordinates, direct recomputation agrees with the tensor transformation law to 1.08×10^{-14}, while contraction with the transformed fourth noise moment is invariant to 2.36×10^{-15}.

The quartic term is not universally sufficient. We therefore compute the zero-error Taylor jet through order thirty by block-matrix exponentials and bound the uncomputed tail analytically. For every declared path, a strict Krawczyk inclusion certifies a locally unique root exactly matched in projective output state and first projective response inside its transverse phase box. Direct 192-bit outward-rounded Arb propagation of these boxes certifies all 66 frozen finite-error pairwise orderings. The order-thirty jet propagated over the same boxes separately certifies 52/66 pairs without a reversed comparison. In the preceding serialized-control formal audit, the quartic term alone certifies 34/66 pairs.

The result is a formal exact-root ordering theorem for the serialized two-atom model, not PASQAL hardware evidence.

1 Introduction

Pulse-level control of programmable neutral-atom arrays provides direct access to Rabi amplitude, phase, detuning, geometry, and Rydberg-mediated interactions [1, 2, 3]. This flexibility also creates a basic identification problem. Two implementations may realize the same ideal state and have the same leading response to calibrated errors, yet behave differently when the errors are finite. Endpoint agreement alone does not identify an implementation, and even first-order robustness data may leave a nontrivial family of controls.

Composite-pulse and robust-control methods commonly cancel or suppress leading error terms [4, 5, 6]. The question asked here is different:

After the ideal output state and the complete first-order output-state response have been matched, what local object predicts the residual finite-error ordering inside the remaining implementation fibre?

We address the question in four stages, corresponding to the four-layer theorem structure below.

1. We construct phase schedules with a common endpoint and common first-order state response to amplitude, detuning, and Rydberg-interaction errors.

2. We test a low-order quartic response scalar, $\mathcal{G}_4$, as a prospective geometric predictor, using a protocol-frozen ranking followed by held-out finite-error evaluation.

3. We promote $\mathcal{G}_4$ to a coordinate-covariant fourth-order tensor whose contraction with a physical noise moment is invariant.

4. We establish strict Krawczyk inclusions for locally unique roots exactly matched in projective output state and first projective response, then produce outward-rounded Arb ball certificates of finite-error ordering for the frozen twelve-path cohort.

The central outcome is deliberately narrower than a universal robustness law. The fourth-order term is often predictive and can certify a substantial subset of pairwise comparisons, but close paths require higher derivatives. For the serialized model studied here, the exact-root direct Arb certificate establishes the complete finite-error ordering, while the order-thirty local jet over the same boxes certifies 52/66 pairs with zero reversals. This distinguishes four statements that are often conflated:

$$\begin{aligned}\text{low-order correlation} &\neq \text{partial geometric certification}\\ &\neq \text{bounded high-order reconstruction}\\ &\neq \text{exact-root finite-radius certification.}\end{aligned}$$

1.1 Contributions

The main contributions of this work are:

- A prospective geometric predictor, $\mathcal{G}_4$, achieving Spearman correlation 0.996992 and top-path correctness on a protocol-frozen prospective twenty-path cohort.

- A coordinate-covariant fourth-order response tensor whose contraction with the physical fourth noise moment is invariant under linear error reparameterization.

- Strict Krawczyk inclusions for 12/12 declared controls, establishing locally unique roots exactly matched in projective output state and first projective response inside transverse phase boxes.

- An exact-root finite-error ordering theorem: all 66 unordered path pairs are certified by outward-rounded interval propagation, with the frozen order preserved.

- A mechanism certificate: order-thirty jet propagation over the same Krawczyk boxes certifies 52/66 pairs with no reversed comparison.

2 Two-atom control model

2.1 Hamiltonian

We use the ordered basis

$$\{|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle\}.$$

Let

$$X_\Sigma = X \otimes I + I \otimes X, \qquad Y_\Sigma = Y \otimes I + I \otimes Y,$$

$$N_\Sigma = n \otimes I + I \otimes n, \qquad N_{rr} = n \otimes n,$$

where $n = |r\rangle\langle r|$. During segment j, the Hamiltonian is

$$\begin{aligned} H_j(\boldsymbol{\epsilon}) = &\frac{\Omega}{2}(1+\epsilon_\Omega)\left(\cos\phi_j\, X_\Sigma + \sin\phi_j\, Y_\Sigma\right) \\ &- \Omega\,\epsilon_\Delta N_\Sigma + V(1+\epsilon_V)N_{rr}. \end{aligned} \tag{1}$$

The dimensionless error vector is

$$\boldsymbol{\epsilon} = (\epsilon_\Omega, \epsilon_\Delta, \epsilon_V).$$

The nominal parameters are

$$\Omega = 2\pi \text{ rad}/\mu\text{s}, \qquad V = 4\Omega,$$

with 24 segments of duration

$$\tau = 0.1\ \mu\text{s}, \qquad T = 24\tau = 2.4\ \mu\text{s}.$$

The nominal interaction can be represented as $V = C_6/R^6$; using the Pulser DigitalAnalogDevice coefficient in the accompanying implementation gives a nominal separation of approximately 7.744 μm.

For a control path

$$\gamma = (\phi_1, \ldots, \phi_{24}) \in \mathbb{T}^{24},$$

the output state is

$$|\psi_\gamma(\boldsymbol{\epsilon})\rangle = \prod_{j=24}^{1} \exp[-\mathrm{i}\tau H_j(\boldsymbol{\epsilon})]|gg\rangle. \tag{2}$$

All primary calculations use exact matrix exponentials for the piecewise constant, finite-dimensional model. Pulser/QutipEmulator was used in an earlier two-atom cross-check, but the formal certificate in this work is independent of Pulser and does not invoke a cloud backend.

2.2 Loss and held-out error law

Let γ_{ref} denote the reference schedule and define

$$|\psi_{\text{ref}}\rangle = |\psi_{\gamma_{\text{ref}}}(\mathbf{0})\rangle.$$

The phase alignment for path γ uses

$$c_\gamma = \frac{\overline{\langle\psi_{\text{ref}}|\psi_\gamma(\mathbf{0})\rangle}}{|\langle\psi_{\text{ref}}|\psi_\gamma(\mathbf{0})\rangle|}. \tag{3}$$

The state infidelity is

$$\mathcal{I}_\gamma(\boldsymbol{\epsilon}) = 1 - |\langle\psi_{\text{ref}}|\psi_\gamma(\boldsymbol{\epsilon})\rangle|^2. \tag{4}$$

The held-out error design contains the six signed axial errors

$$\mathcal{E} = \{\pm 0.06\, e_\Omega,\ \pm 0.04\, e_\Delta,\ \pm 0.05\, e_V\}. \tag{5}$$

The primary performance functional is the mean

$$\overline{\mathcal{I}}_\gamma = \frac{1}{6} \sum_{\boldsymbol{\epsilon} \in \mathcal{E}} \mathcal{I}_\gamma(\boldsymbol{\epsilon}). \tag{6}$$

Worst-case infidelity is retained as a secondary diagnostic. The present predictors do not reliably rank the worst case, and no worst-case theorem is claimed.

3 Matched implementation fibre

3.1 Endpoint and first-order constraints

At zero error, remove the unobservable global phase by the horizontal projector

$$P_\perp = I - |\psi_0\rangle\langle\psi_0|.$$

Define the horizontal first-order responses

$$J_{\gamma,a} = P_\perp \frac{\partial}{\partial \epsilon_a} |\psi_\gamma(\boldsymbol{\epsilon})\rangle \bigg|_{\boldsymbol{\epsilon}=0}, \qquad a \in \{\Omega, \Delta, V\}. \tag{7}$$

Definition 1 (Numerically matched implementation fibre)**.** *For a declared tolerance η, the numerical matched fibre $\mathcal{F}_\eta$ consists of controls γ satisfying*

$$\begin{aligned} \mathcal{I}_\gamma(\mathbf{0}) &\le \eta, \\ \|J_{\gamma,a} - J_{\mathrm{ref},a}\| &\le \eta, \qquad a \in \{\Omega, \Delta, V\}. \end{aligned} \tag{8}$$

The actual optimizer retains the full phase-aligned complex output state and the three full phase-aligned complex state tangents. Redundant real components are retained in the residual vector, while singular-value decomposition determines the independent rank.

3.2 Dimension count and generation

The control has 24 phase variables. At the reference solution, the numerical constraint Jacobian has rank

$$\operatorname{rank} D\mathcal{C}(\gamma_{\mathrm{ref}}) = 16,$$

leaving an eight-dimensional numerical null space that organizes the candidate implementation fibre. Candidate paths are generated by:

1. drawing a seeded direction in the numerical null space;
2. applying a finite displacement;
3. projecting back to the constraint set by nonlinear least squares;
4. rejecting candidates whose constraint residual or pairwise phase distance fails a predeclared gate.

No held-out finite-error outcome is evaluated during candidate generation. The maximum constraint residual in the reported runs is of order 10^{-11}.

Remark 1 (Global fibre versus local certified roots)**.** *The numerical rank and residual establish a highly accurate computational fibre, suggesting a locally eight-dimensional numerical implementation fibre, not the exact existence of a global eight-dimensional solution manifold. A complete global fibre theorem would require a square transverse formulation and a global interval-Newton or Krawczyk proof. The present work instead establishes 12 locally unique roots exactly matched in projective output state and first projective response via strict Krawczyk inclusions in transverse boxes. The global fibre structure remains open.*

4 L1: Prospective geometric prediction

4.1 Symmetric local response

Introduce normalized error coordinates $z \in \mathbb{R}^3$ through

$$\boldsymbol{\epsilon} = Dz, \qquad D = \mathrm{diag}(0.06, 0.04, 0.05).$$

For a direction $v \in \mathbb{R}^3$, define the symmetric response

$$S_\gamma(t, v) = \frac{\mathcal{I}_\gamma(tDv) + \mathcal{I}_\gamma(-tDv)}{2}. \tag{9}$$

Analyticity of the finite matrix exponential gives

$$S_\gamma(t, v) = q_{2,\gamma}(v)t^2 + q_{4,\gamma}(v)t^4 + q_{6,\gamma}(v)t^6 + \cdots . \tag{10}$$

Odd powers cancel by construction.

First-order output-state matching fixes the leading quadratic loss to numerical precision. Across the validation cohorts, the relative spread of the averaged quadratic coefficient is approximately 10^{-7}. The first useful discriminator is therefore the quartic coefficient.

4.2 Prospective quartic ranking

4.2.1 Discovery and validation separation

The initial four named controls were used only for discovery and orientation. A twelve-path cohort was then used to compare candidate formulas. The smallest sufficient rule on that training cohort was

$$\text{smaller } \mathcal{G}_4 \quad \Longrightarrow \quad \text{smaller mean finite-error infidelity.}$$

The addition of the sixth-order coefficient did not improve the training rank correlation, so the quartic rule was frozen.

A new twenty-path cohort was generated with a different seed. For each candidate, a ranking certificate containing phases, local coefficients, and predicted order was written and SHA-256 hashed before the held-out outcome evaluator was enabled. The predeclared primary gate was

$$\rho_{\text{Spearman}} \geq 0.95,$$

together with correct top-path identification.

4.2.2 Quartic validation result

The twenty-path Colab run gave

$$\rho_{\text{Spearman}}\left(-\mathcal{G}_4, -\overline{\mathcal{I}}\right) = 0.996992, \qquad p = 1.89 \times 10^{-21}. \tag{11}$$

The predicted and observed top path agreed. Only one adjacent pair was inverted in the complete ranking. Two local coefficient-extraction windows had rank correlation one, and the maximum local fit residual was 1.15×10^{-10}.

Subsequent cohorts show that $\mathcal{G}_4$ is not universally sufficient. Depending on the generated matched paths, prospective quartic Spearman correlations ranged from approximately 0.85 to 1.00. This variation motivated the bounded higher-order analysis rather than post hoc replacement of the quartic formula.

Remark 2 (Scope of L1)**.** *The quartic predictor is a geometric predictor, not a universal theorem. It strongly organizes mean finite-error performance but does not alone certify all pairwise comparisons.*

5 L2: Covariant fourth-order tensor

5.1 Quartic tensor

There exists a symmetric fourth-order tensor $A^{(4)}_\gamma$ such that

$$q_{4,\gamma}(v) = A^{(4)}_{\gamma,ijkl} v_i v_j v_k v_l. \tag{12}$$

In three dimensions a symmetric fourth-order tensor has

$$\binom{3+4-1}{4} = 15$$

independent components. We recover them from 36 seeded directions. The directional Taylor coefficients are computed directly at $t = 0$ through order six by block-matrix exponentials, rather than by fitting nonzero-radius samples.

For the six-axis law in Eq. (5), define the fourth moment

$$M^{(4)}_{ijkl} = \mathbb{E}[z_i z_j z_k z_l].$$

The scalar quartic predictor is

$$\mathcal{G}_4(\gamma; M^{(4)}) = A^{(4)}_{\gamma,ijkl} M^{(4)}_{ijkl}. \tag{13}$$

For the equally weighted signed axes,

$$\mathcal{G}_4 = \frac{1}{3}\left(A^{(4)}_{\Omega\Omega\Omega\Omega} + A^{(4)}_{\Delta\Delta\Delta\Delta} + A^{(4)}_{VVVV}\right).$$

5.2 Coordinate covariance

Proposition 1 (Covariance and invariant contraction)**.** *Let $z = Ry$ for an invertible real matrix R. Then*

$$A^{(4)}_{y,abcd} = A^{(4)}_{z,ijkl} R_{ia} R_{jb} R_{kc} R_{ld}, \tag{14}$$
$$M^{(4)}_{y,abcd} = (R^{-1})_{ai}(R^{-1})_{bj}(R^{-1})_{ck}(R^{-1})_{dl} M^{(4)}_{z,ijkl}, \tag{15}$$

and

$$A^{(4)}_y : M^{(4)}_y = A^{(4)}_z : M^{(4)}_z. \tag{16}$$

Proof. Equation (14) follows by substituting $z = Ry$ into the quartic form. Equation (15) follows from $y = R^{-1}z$. Contracting all four indices cancels each R with the corresponding R^{-1}, proving Eq. (16). □

This proposition is algebraic. The numerical audit additionally recomputes the tensor directly in the transformed coordinates. For a nonorthogonal map of condition number approximately 1.82, the direct-refit relative error is 1.08×10^{-14} and the maximum invariant-contraction relative error is 2.36×10^{-15}.

The covariant object is not the scalar $\mathcal{G}_4$ in isolation. It is the fourth-order response tensor $A^{(4)}$ together with the physical fourth noise moment $M^{(4)}$. Their contraction is independent of a linear reparameterization of error coordinates. This avoids interpreting a coordinate-specific axis average as an intrinsic curvature.

The tensor should also not be identified automatically with Riemann curvature. It is a quartic susceptibility of the endpoint loss—a component of the fourth jet of the endpoint map composed with infidelity. Establishing a relation to a canonical connection or curvature on a control quotient would require additional structure.

For a noise moment M, the low-cost screening problem is

$$\gamma^\star \in \arg\min_{\gamma \in \mathcal{F}_\eta} A_\gamma^{(4)} : M^{(4)}.$$

Pairs that fail the quartic gap test can be escalated adaptively to sixth, eighth, or higher jet order until their intervals separate. This suggests a certified branch-and-bound control selector: compute low-order local tensors for all candidates, refine only ambiguous pairs, and reserve direct finite-error simulation or hardware evaluation for model validation rather than exhaustive ranking.

The gap-dependent quartic certification criterion is:

$$|\mathcal{G}_{4,a} - \mathcal{G}_{4,b}| > B_a^{(\geq 6)} + B_b^{(\geq 6)} \quad \Longrightarrow \quad \text{quartic ordering is certified.} \tag{17}$$

6 L3: Certified exact projective matched roots via Krawczyk inclusions

The numerical paths satisfy the endpoint and first-order matching constraints to order 10^{-11}. To move from numerical candidates to exact mathematical roots, we formulate the constraint map $\mathcal{C} : \mathbb{R}^{24} \to \mathbb{R}^{16}$ (after eliminating redundant real components) and decompose the phase space into 16 transverse variables x and 8 fibre variables y. For each declared numerical path γ_k, we fix its fibre coordinates y_k and consider the reduced map

$$F_k(x) = \mathcal{C}(x, y_k).$$

The projective chart denominator is formally verified to exclude zero on every accepted box, ensuring the coordinate chart is valid throughout the Krawczyk computation.

Theorem 1 (Krawczyk inclusions for 12 declared controls)**.** *For each of the twelve declared controls, there exists a transverse phase box*

$$X_k \subset \mathbb{R}^{16}$$

such that the Krawczyk operator satisfies

$$K_k(X_k) \subset \operatorname{int}(X_k).$$

Consequently, each box X_k contains a locally unique exact root x_k^* of $F_k(x) = 0$, i.e., a control $\gamma_k^* = (x_k^*, y_k)$ whose root is exactly matched in projective output state and first projective response.

Computer-assisted proof. A numerical approximate inverse of the midpoint Jacobian is used as the point preconditioner. The Jacobian over each full box is then enclosed with 192-bit Arb arithmetic, and the resulting Krawczyk image is verified to lie strictly inside the box. The projective chart denominator is formally verified to exclude zero on every accepted box. For all twelve boxes, the inclusion $K(X) \subset \operatorname{int}(X)$ is verified. The protocol hash is

`a19d01ec77a783be4e3858dd92ca080a`
`a9fde2b09b77541b0e1ad7f392ff3124`

and the certificate hash is

`fc3ad3823597bd5281b7b9292f227f72`
`3c2b3a9a8aa4e42c9f0353921b4d8fec`.

By the Krawczyk theorem [10, 9], each box contains a locally unique root. □

Remark 3 (Scope of L3)**.** *The theorem establishes local uniqueness inside each declared transverse box. It does not certify global uniqueness on the entire fibre, nor does it prove that every point on the numerical fibre has an exact representative. The global fibre structure remains open.*

7 L4: Exact-root finite-error ordering

7.1 Direct Arb propagation of Krawczyk boxes

The following theorem is the main result of this work. Its proof uses only the certified Krawczyk boxes and direct outward-rounded propagation of the finite-error functional. No Taylor truncation, Cauchy extraction, alias enclosure, or local-jet reconstruction is used.

Theorem 2 (Exact-root finite-error ordering)**.** *For each of the twelve declared controls, let X_k be the transverse phase box satisfying a strict Krawczyk inclusion from Theorem 1. Direct outward-rounded Arb propagation of X_k through the declared six-point finite-error functional produces twelve pairwise ordered performance intervals. All* 66 *unordered path pairs are disjoint and agree with the prospectively frozen ordering. Therefore the locally unique roots exactly matched in projective output state and first projective response inside the boxes obey the complete frozen finite-error ordering.*

Computer-assisted proof. By Theorem 1, the locally unique exact projective matched control γ_k^* belongs to the certified phase box X_k.

For every box X_k, the six declared finite-error Hamiltonians are evaluated directly using 192-bit outward-rounded Arb/ACB arithmetic. Every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. Denote the resulting real performance interval by B_k. Therefore

$$\overline{\mathcal{I}}(\gamma_k^*) \in B_k.$$

For every frozen ordered pair $a \prec b$, direct interval comparison verifies

$$\sup B_a < \inf B_b.$$

All $\binom{12}{2} = 66$ comparisons pass and have the orientation specified by the frozen ordering certificate. Hence

$$\overline{\mathcal{I}}(\gamma_a^*) < \overline{\mathcal{I}}(\gamma_b^*)$$

for every frozen ordered pair. No Taylor truncation or local-jet reconstruction is used in this direct certification.

The predicted ordering was fixed by the predecessor zero-point ordering certificate before the exact-root direct finite-error propagation was evaluated. The final exact-root certificate, containing the formal direct intervals and all pairwise comparisons, was written and hashed after the propagation.

The exact-root ordering protocol hash is

```
f51ecfd2e4572f271e276cae18697005
baa3f0ec3f3ab94f98f919cd50c3f844
```

and the certificate hash is

```
03c4db75cb80a149ce1a61e11c227039
f9bd79c3f1304b3defa311a491ea2d1e.
```

□

Remark 4 (Scope of Theorem 2)**.** *The theorem is conditional on the serialized-decimal finite Hamiltonian model, the declared transverse boxes, and the six-axis mean error functional. It does not certify model discrepancy, global fibre topology, many-body scaling, or PASQAL hardware. It also does not prove a global eight-dimensional exact fibre theorem; it proves finite-error ordering for the locally unique exact roots inside the declared boxes.*

Remark 5 (Integrity check)**.** *In all reported runs, every ordinary held-out floating-point outcome is verified to lie inside its corresponding formal ball. This serves as an integrity diagnostic and is not part of the theorem proof.*

7.2 Order-thirty mechanism certificate

The following proposition provides a separate, independent mechanism certificate using the zero-point jet. Unlike the main theorem, it uses Taylor coefficients, Cauchy extraction, alias enclosures, and Dyson tail bounds. It is not part of the main theorem proof.

The floating-point precursor obtains the order-thirty jet by block-matrix exponentials. The formal phase-box certificate used in Proposition 2 instead extracts the coefficients with a 64-point outward-rounded Cauchy transform.

Proposition 2 (Order-thirty mechanism certificate)**.** *Propagation of the zero-point jet through order thirty over the twelve Krawczyk phase boxes, including the formal Cauchy-alias and analytic order-32 tail enclosures, certifies* $52/66$ *frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified.*

Computer-assisted proof. Amplitude coefficients are extracted by a 64-point outward-rounded Cauchy discrete Fourier transform on the complex unit circle:

$$\widehat{g}_n = \frac{1}{64}\sum_{k=0}^{63} g(\zeta_k)\zeta_k^{-n}, \qquad \zeta_k = e^{2\pi \mathrm{i}k/64}. \tag{18}$$

The discrete transform aliases coefficients $g_{n+64\ell}$. A path-uniform Dyson bound encloses this alias. The implemented enclosure is 10^{-40}, and the formal gate verifies that

$$e^{K_a}\frac{K_a^{64}}{64!} < 10^{-40}$$

for every physical error axis, where K_a is the integrated spectral norm of the Hamiltonian perturbation along axis a:

$$K_a = \int_0^T \|\partial_t H_a(t)\|_2 \,\mathrm{d}t.$$

Fidelity coefficients are formed by ball convolution,

$$F_n = \sum_{r=0}^{n} \overline{g_r} g_{n-r},$$

and the infidelity coefficients follow from $1 - F$. The analytic even Dyson tail after order thirty is enclosed by 2×10^{-11}.

Propagating the resulting order-thirty jet over the twelve Krawczyk phase boxes yields performance intervals. Pairwise comparison of the resulting order-thirty intervals certifies $52/66$ frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified. The unresolved pairs arise from dependency and wrapping in the phase-box propagation of the jet enclosure. They are not reversed physical comparisons and do not indicate an omitted-tail failure. □

8 Results

The prospective G_4 prediction audit uses a held-out twenty-path cohort. The Krawczyk, exact-root order-30, and direct exact-root certificates use the separate frozen twelve-path cohort declared in the v0.3.1 formal artifacts.

The sequence of results supports the following hierarchy.

1. **Implementation memory:** endpoint and complete first-order output-state matching do not numerically determine finite-error performance.

Table 1: Representative prospective and formal audits. "Pairs" denotes certified unordered comparisons. Candidate sets differ between stages and between numerical environments; every stage freezes its protocol and ranking before held-out evaluation. The direct exact-root ordering is the main theorem.

Audit	Paths	Spearman ρ	Certified pairs	Status
Prospective G_4 prediction	20	0.996992	–	pass
Covariant fourth-order tensor	12	1.000000	–	pass
Formal quartic-only certificate	12	–	34/66	partial
Frozen-point order-30 certificate	12	1.000000	66/66	pass
Exact-root Krawczyk inclusion	12	–	12/12	pass
Exact-root order-30 mechanism	12	–	52/66	partial
Exact-root direct Arb ordering	12	1.000000	66/66	pass

Table 2: Numerical integrity diagnostics from representative Colab runs.

Diagnostic	Value
Fourth-tensor independent components	15
Maximum tensor recovery relative residual	4.16×10^{-15}
Direct coordinate-refit relative error	1.08×10^{-14}
Invariant-contraction relative error	2.36×10^{-15}
Quadratic-coefficient relative spread	1.33×10^{-7}
Arb precision	192 bits
Cauchy points	64
Taylor order	30
Formal quartic pair coverage	$34/66 = 51.52\%$
Exact-root order-30 mechanism coverage	$52/66 = 78.79\%$
Exact-root direct Arb ordering coverage	$66/66 = 100\%$

2. **Low-order organization:** the covariant quartic contraction $\mathcal{G}_4$ strongly organizes mean performance and formally certifies a substantial subset of path pairs (34/66 at the formal ball level).

3. **Exact-root certification:** strict Krawczyk inclusions establish 12 locally unique roots exactly matched in projective output state and first projective response.

4. **High-order mechanism:** the order-thirty jet propagated over the Krawczyk boxes certifies 52/66 pairs with zero reversals (Proposition 2).

5. **Complete ordering:** direct 192-bit outward-rounded Arb propagation of the Krawczyk boxes certifies all 66 pairs with the frozen order preserved (Theorem 2).

9 Discussion

9.1 Why the quartic term is useful but incomplete

First-order matching makes the quadratic loss nearly common. This promotes the quartic term to the first visible discriminator. It explains why $\mathcal{G}_4$ often gives almost perfect rankings. However, if two quartic coefficients are close, the sixth and higher terms can reverse their order.

The formal intervals expose this distinction directly: 34 pairs are quartic-certified, while the remaining pairs require higher jet data or direct propagation.

This is preferable to declaring the quartic rule either universally true or false. The correct statement is gap-dependent as in Eq. (17).

9.2 Order-30 mechanism versus direct propagation

The order-30 jet propagated over the Krawczyk boxes certifies 52/66 pairs with zero reversals (Proposition 2). The remaining 14 pairs are not physically reversed—they are unresolved due to dependency and wrapping in the phase-box propagation of the jet enclosure. Direct Arb propagation of the same boxes certifies all 66 pairs (Theorem 2), demonstrating that the unresolved cases are an artefact of the jet propagation and interval wrapping, not a physical failure of the frozen ordering.

Remark 6 (On 52/66 vs 66/66)**.** *The* 52/66 *and* 66/66 *numbers arise from different certificates and different propagation levels. They should not be conflated. The* 52/66 *mechanism certificate shows that the order-thirty jet already resolves a substantial fraction without reversal. The* 66/66 *direct certificate is the main theorem.*

9.3 Relation to analyticity

An analytic finite-dimensional propagator is determined by its full local germ, so the existence of some sufficiently high reconstructing order is not by itself surprising. The nontrivial computational content is quantitative:

- a finite order, 30, suffices to certify a substantial fraction (52/66) of exact-root comparisons with zero reversals;
- the omitted tail is explicitly bounded and does not contribute to the unresolved cases (which arise from wrapping, not truncation);
- direct outward-rounded propagation closes the remaining cases;
- a much lower fourth-order object already certifies a substantial fraction (34/66) of comparisons in the preceding serialized-control audit.

Thus the result is not that "analytic functions have Taylor series," but that a finite, auditable local resource closes a nontrivial control-ranking problem when combined with exact-root certification.

10 Limitations and next theorem

The present work has five explicit limitations.

1. **Global fibre structure.** Twelve locally unique roots exactly matched in projective output state and first projective response are certified in transverse boxes. The global eight-dimensional solution manifold structure remains unproved.
2. **Finite-dimensional model.** Spontaneous emission, Doppler effects, laser phase noise, position fluctuations, leakage levels, and waveform filtering are absent.
3. **Two atoms.** No many-body scaling theorem is claimed.
4. **Mean axial error law.** The primary result concerns the declared six-axis mean. Worst-case ranking is not supported by the present scalar.

5. **No QPU evidence.** The calculations are local and require no PASQAL credentials. They do not constitute PASQAL Cloud, FRESNEL, or physical-QPU evidence.

The most accurate limitations sentence is:

> The certificate is formal for the serialized finite-dimensional two-atom Hamiltonian, the declared transverse boxes, and the six-axis mean error functional. It does not certify model discrepancy, global fibre topology, many-body scaling, or PASQAL hardware.

The next mathematical step is a validated implicit-function theorem or a parameter-dependent Krawczyk atlas: treat the eight fibre coordinates as parameters and certify a unique transverse solution varying over a declared fibre neighbourhood. The target mathematical structure is

$$F(x, y) = 0, \qquad x \in \mathbb{R}^{16},\ y \in \mathbb{R}^{8},$$

and the goal is to prove that a unique transverse solution $x = x(y)$ exists and is regular over a declared y-region, not to prove a single global isolated root.

11 Reproducibility

The principal standalone scripts are:

- `pasqal_two_atom_G4_standalone_colab.py`;
- `pasqal_L3_L4_standalone_colab.py`;
- `pasqal_L4_order30_standalone_colab.py`;
- `pasqal_L4_formal_arb_standalone_colab.py`;
- `pasqal_L4_exact_fibre_krawczyk_standalone_colab_v1_2.py`;
- `pasqal_L4_exact_root_ordering_standalone_colab.py`.
- `pasqal_L4_reproducible_certificate_v1_3_colab.py`.

The v1.3 standalone freezes the twelve-path cohort, transverse bases, and point preconditioners, fixes the common Krawczyk radius at 3×10^{-12}, and excludes runtime metadata from every hashed proof object. Two complete formal executions produce byte-identical protocol, certificate, and report files. The frozen cohort hash is

```
9942af610c1e9499a10e989e4ae069e3
912d58eec4fd3ad1d3d39947c232c356.
```

The deterministic protocol and certificate hashes are:

- Krawczyk protocol:

```
a19d01ec77a783be4e3858dd92ca080a
a9fde2b09b77541b0e1ad7f392ff3124
```

- Krawczyk certificate:

```
fc3ad3823597bd5281b7b9292f227f72
3c2b3a9a8aa4e42c9f0353921b4d8fec
```

- Exact-root ordering protocol:

  `f51ecfd2e4572f271e276cae18697005`
  `baa3f0ec3f3ab94f98f919cd50c3f844`

- Exact-root ordering certificate:

  `03c4db75cb80a149ce1a61e11c227039`
  `f9bd79c3f1304b3defa311a491ea2d1e`

The formal standalone script pins `python-flint==0.8.0`. NumPy and SciPy provide numerical diagnostics, while the v1.3 proof consumes the frozen decimal cohort and uses Arb/ACB for the formal interval certificate.

Code and data availability

All source code, frozen numerical inputs, outward-rounded interval certificates, verification utilities, and manuscript source are available in the companion repository *Quantum Control Geometry*:

`https://github.com/papasop/quantum-control-geometry`

The formal computational artifacts used in this work are fixed in release `v0.3.1` at commit

`284974c9f6b952f4e114c8c5bdc9c2c299c4065c`.

The synchronized manuscript source is maintained on the repository's `main` branch. The archival DOI will be inserted after deposition.

12 Conclusion

We establish two complementary results on distinct frozen cohorts. On a prospective twenty-path cohort, the zero-point quartic geometric scalar strongly predicts held-out finite-error performance. On a separate twelve-path formal cohort, direct outward-rounded interval propagation certifies all 66 pairwise finite-error orderings at locally unique roots exactly matched in projective output state and first projective response. We have separated finite-error robustness inside a numerically matched neutral-atom implementation fibre into a hierarchy of local information. The common endpoint and complete first-order projective state response leave residual path dependence. A coordinate-covariant fourth-order response tensor provides a strong and partially certifying low-order description. Close comparisons require higher derivatives. For the frozen twelve-path serialized model, strict Krawczyk inclusions establish 12 locally unique roots exactly matched in projective output state and first projective response. The zero-error jet through order thirty, combined with rigorous Cauchy-alias and Dyson-tail enclosures, certifies 52/66 exact-root comparisons with zero reversals as a separate mechanism certificate. Direct 192-bit outward-rounded Arb propagation of the Krawczyk boxes certifies all 66 pairwise orderings as the main theorem.

The finite local jet supplies the prospectively frozen geometric ordering and a 52/66 exact-root mechanism certificate, while direct outward-rounded phase-box propagation certifies the complete 66/66 ordering at the locally unique exact projective matched roots. Establishing a global fibre theorem and extending the certificate to open-system, many-atom, and hardware data are the remaining steps.

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